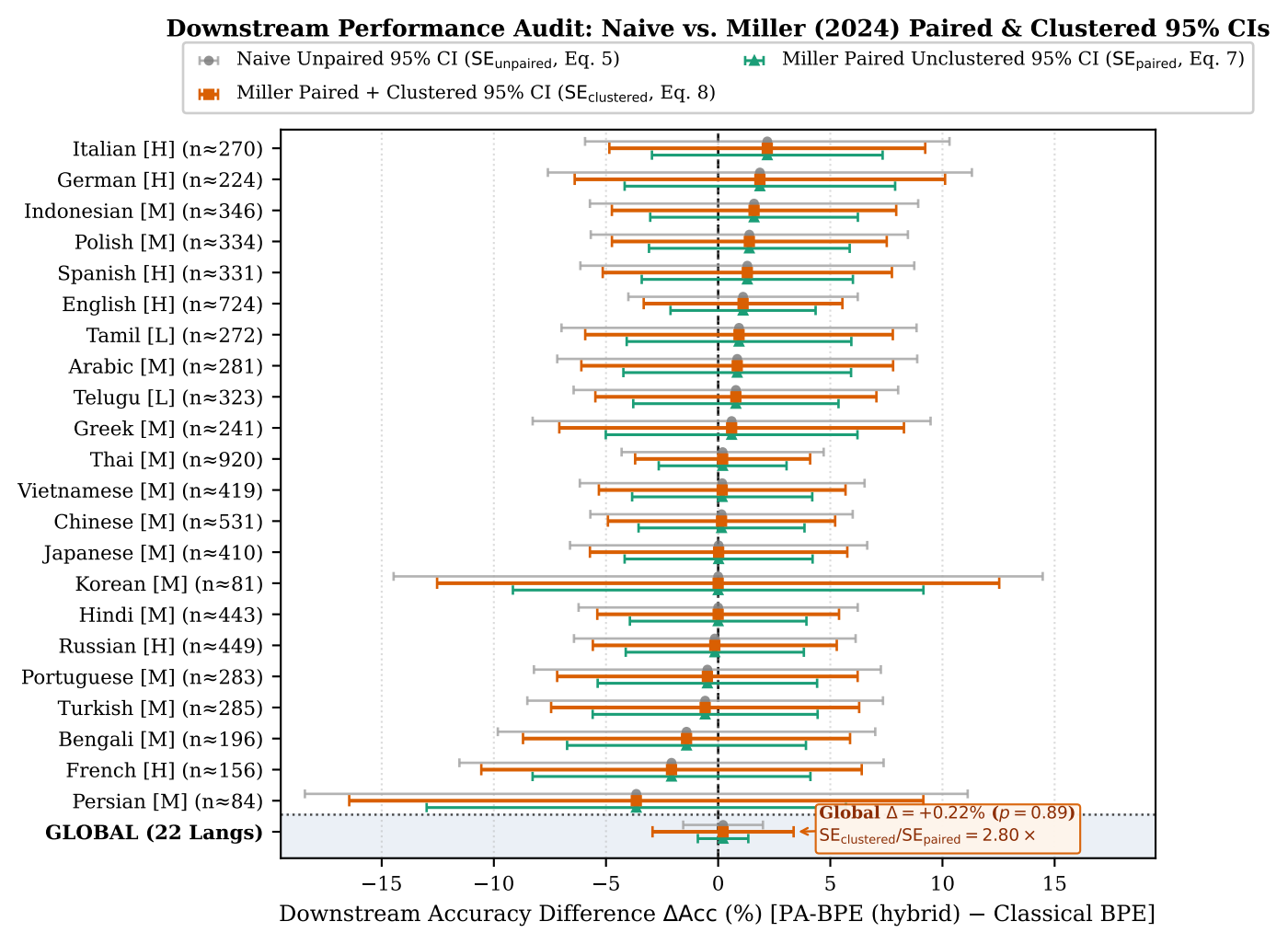

Downstream Performance Audit: Naive vs. Miller (2024) Paired & Clustered 95% CIs
Naive Unpaired 95% CI (SE$_{unpaired}$, Eq. 5)
Miller Paired Unclustered 95% CI (SE$_{paired}$, Eq. 7)
Miller Paired + Clustered 95% CI (SE$_{clustered}$, Eq. 8)
Italian [H] (n≈270)
German [H] (n≈224)
Indonesian [M] (n≈346)
Polish [M] (n≈334)
Spanish [H] (n≈331)
English [H] (n≈724)
Tamil [L] (n≈272)
Arabic [M] (n≈281)
Telugu [L] (n≈323)
Greek [M] (n≈241)
Thai [M] (n≈920)
Vietnamese [M] (n≈419)
Chinese [M] (n≈531)
Japanese [M] (n≈410)
Korean [M] (n≈81)
Hindi [M] (n≈443)
Russian [H] (n≈449)
Portuguese [M] (n≈283)
Turkish [M] (n≈285)
Bengali [M] (n≈196)
French [H] (n≈156)
Persian [M] (n≈84)
GLOBAL (22 Langs)
Global Δ = +0.22% (p = 0.89)
SE$_{clustered}$/SE$_{paired}$ = 2.80 ×
−15
−10
−5
0
5
10
15
Downstream Accuracy Difference ΔAcc (%) [PA-BPE (hybrid) − Classical BPE]